

Sampling contacts from sequence

A figure-first writeup on existing predictors, oracle contacts, MarinFold and useful sampling diversity.

Fixed model: exp277 step 266344, 1.47B parameters, 248.584B raw training tokens. Natural proteins lead: 97 validation + 217 publication test. Designs remain separate. AF2/AF3/Boltz-2 comparators use shared archived MSAs and no templates, with confidence selection.

Predictor generation → cached analysis tables → fast static and interactive rendering. Every numerical figure element maps back to a source CSV row.

What the contact and confidence analyses show

AF2 / AF3 / Boltz-2 GDT-TS: 0.854 / 0.860 / 0.861 on 305 natural proteins. At MSA depth <10: 0.196 / 0.332 / 0.239, with only five proteins.

MarinFold contact R-precision: 0.561 overall; 0.345 at MSA depth <10 versus 0.617 at ≥ 1000 . Only five natural proteins occupy the shallowest tier.

Oracle confidence exceeds every matched random-map control on all 20 preselected proteins. Three diffusion samples per map, identical positive/negative counts and known mask. Random maps are weak negatives, not plausible alternatives.

Top-L contacts raise Helico GDT-TS from 0.150 to 0.504 on 305 matched natural proteins. The paired gain is 0.354 [0.323, 0.387]. All cuts were fixed before test inference.

Useful diversity is the open question

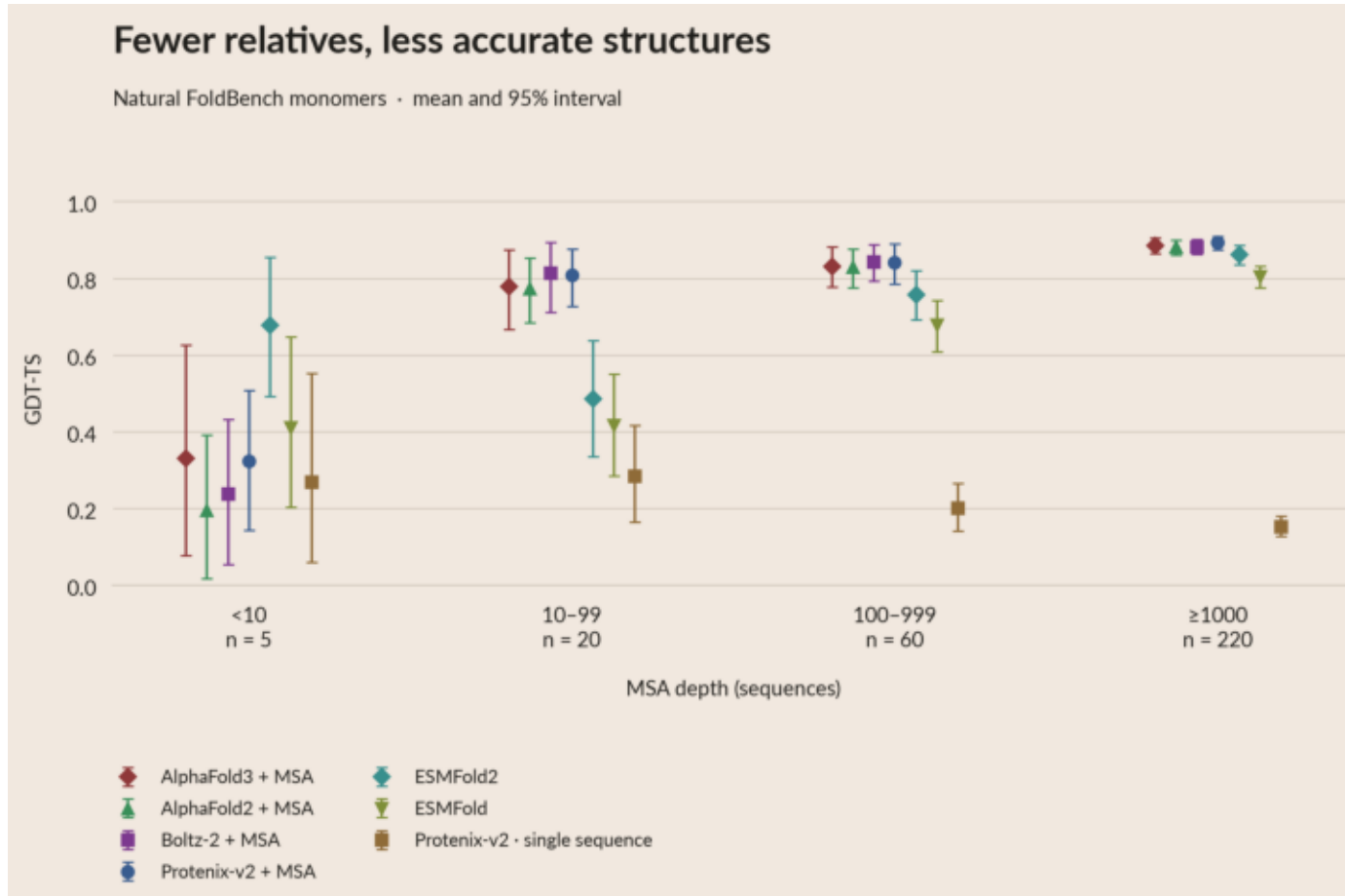
314 proteins $\times$ 100 samples: consensus R-precision 0.561; oracle best individual map 0.528. Paired difference -0.032 , 95% bootstrap interval $[-0.038, -0.025]$.

All 100 maps are distinct for every protein; mean Jaccard 0.277, true-contact union recall 0.943. This is not an absence of contact-set diversity, and it does not directly measure distinct folds.

Next: better whole-map candidates, inference-time search and post-training.

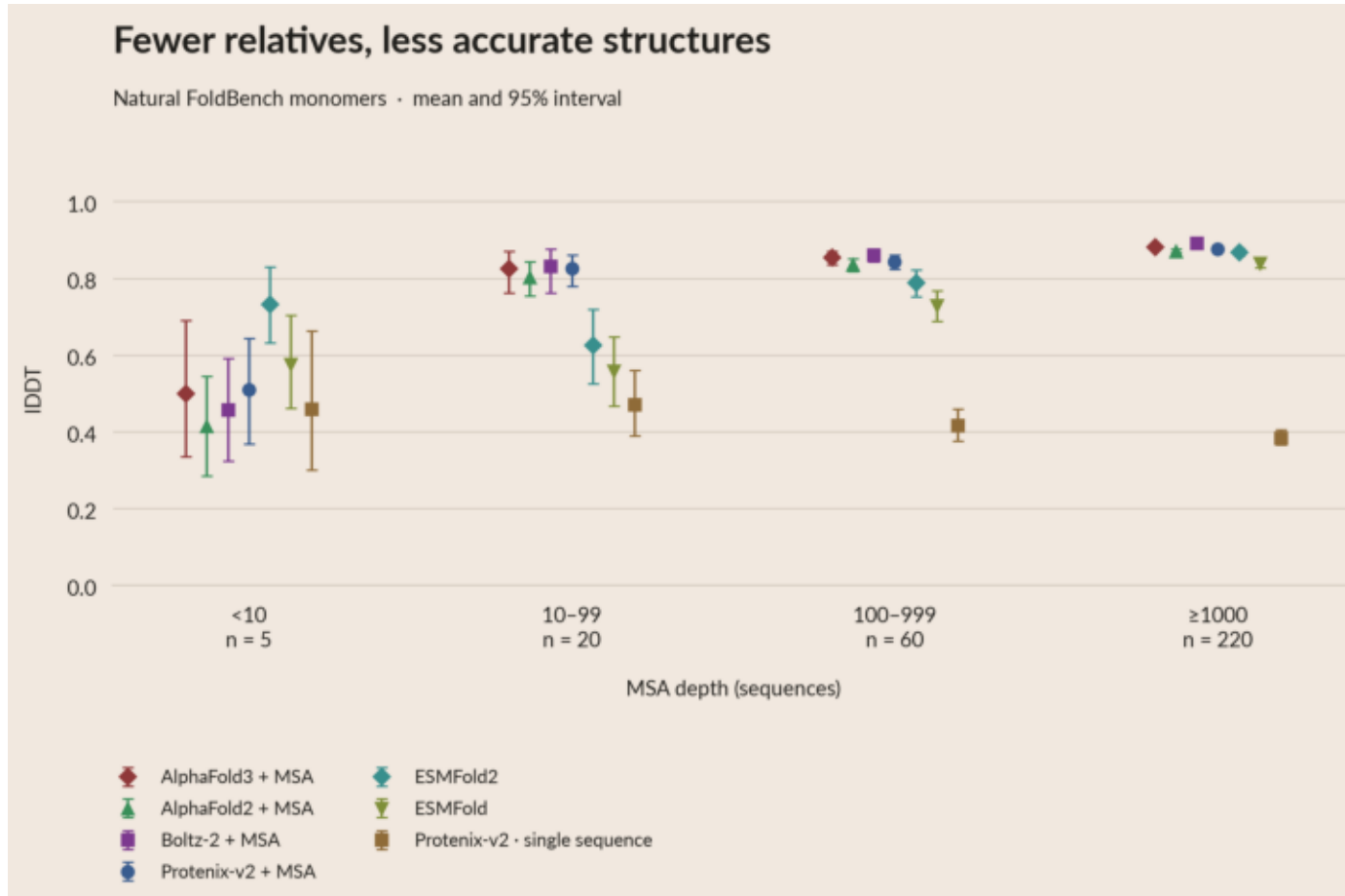
01_predictors.png

Natural FoldBench monomers; the same 305 proteins in every structural arm. Points are protein means; bars are 95% protein-bootstrap intervals. MSA depth counts sequences, including the query, in the alignment used by Protenix-v2 + MSA. AlphaFold2/3 and Boltz-2 use the same archived MSA queries and alignments underlying these depth bins, with templates disabled and protein-chain-only inputs. Confidence selects among five AF2 pTM models (three recycles), 25 AF3 samples (five seeds, ten recycles), or 25 Boltz-2 samples (one seed, ten recycles). These are controlled shared-MSA runs, not the models' full default search pipelines.



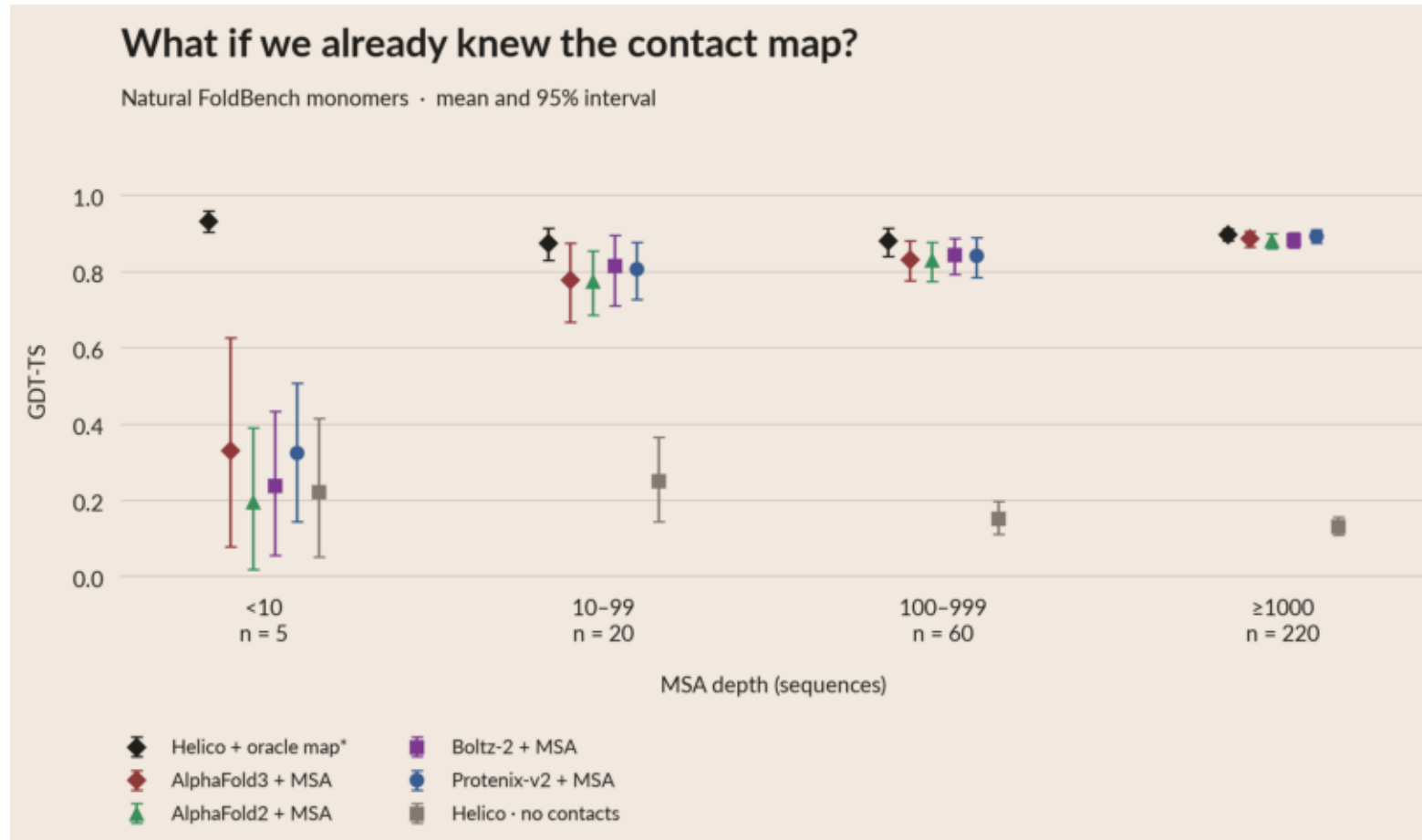
01_predictors_lddt.png

Natural FoldBench monomers; the same 305 proteins in every structural arm. Points are protein means; bars are 95% protein-bootstrap intervals. MSA depth counts sequences, including the query, in the alignment used by Protenix-v2 + MSA. AlphaFold2/3 and Boltz-2 use the same archived MSA queries and alignments underlying these depth bins, with templates disabled and protein-chain-only inputs. Confidence selects among five AF2 pTM models (three recycles), 25 AF3 samples (five seeds, ten recycles), or 25 Boltz-2 samples (one seed, ten recycles). These are controlled shared-MSA runs, not the models' full default search pipelines.



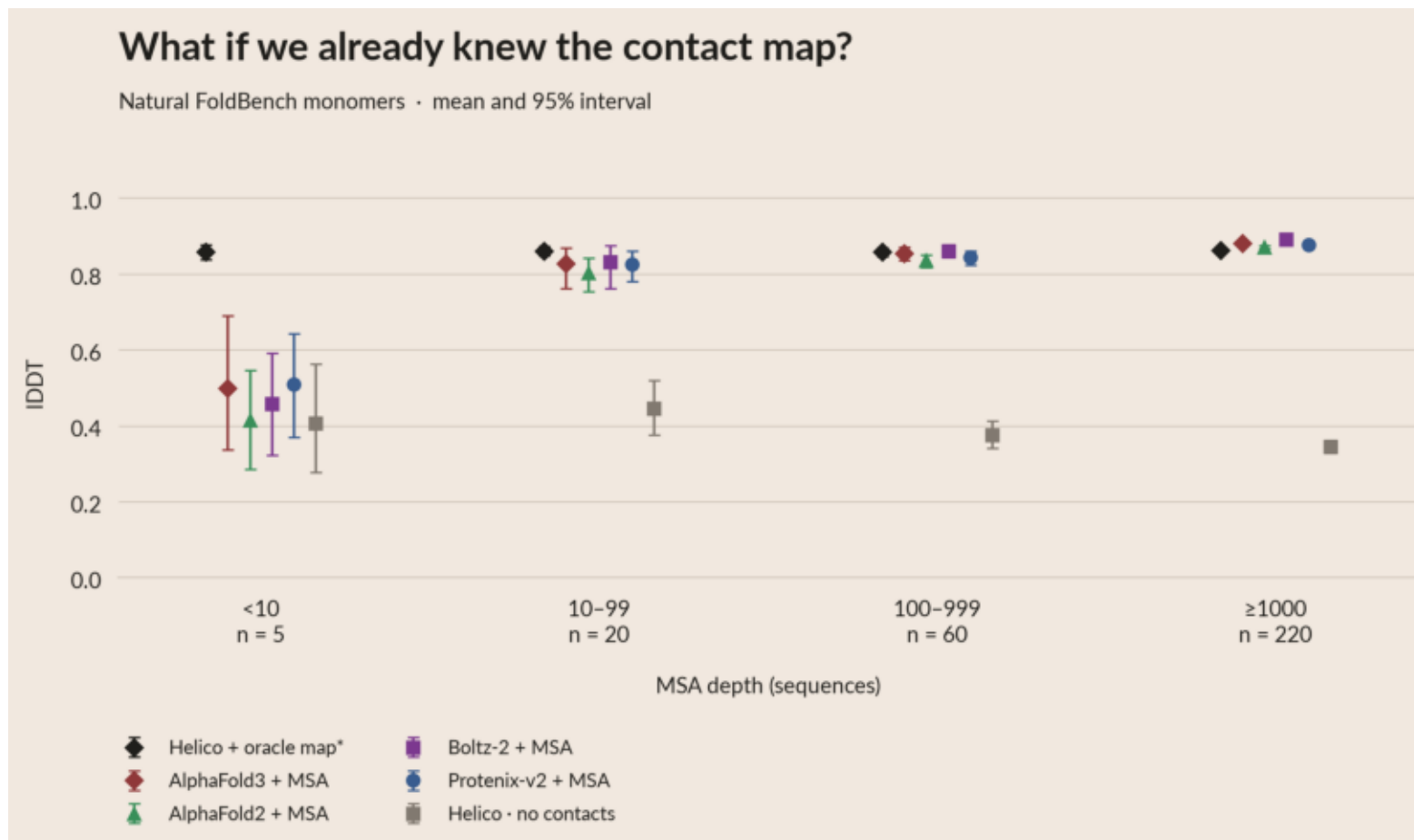
02_oracle.png

*Oracle = ground-truth contacts AND non-contacts supplied to Helico without an MSA. This is an information upper bound, not a competing predictor or a matched-count true-contact experiment. Same 305 natural proteins as Figure 1. AlphaFold2/3 and Boltz-2 use the same archived MSA queries and alignments underlying these depth bins, with templates disabled and protein-chain-only inputs. Confidence selects among five AF2 pTM models (three recycles), 25 AF3 samples (five seeds, ten recycles), or 25 Boltz-2 samples (one seed, ten recycles). These are controlled shared-MSA runs, not the models' full default search pipelines.



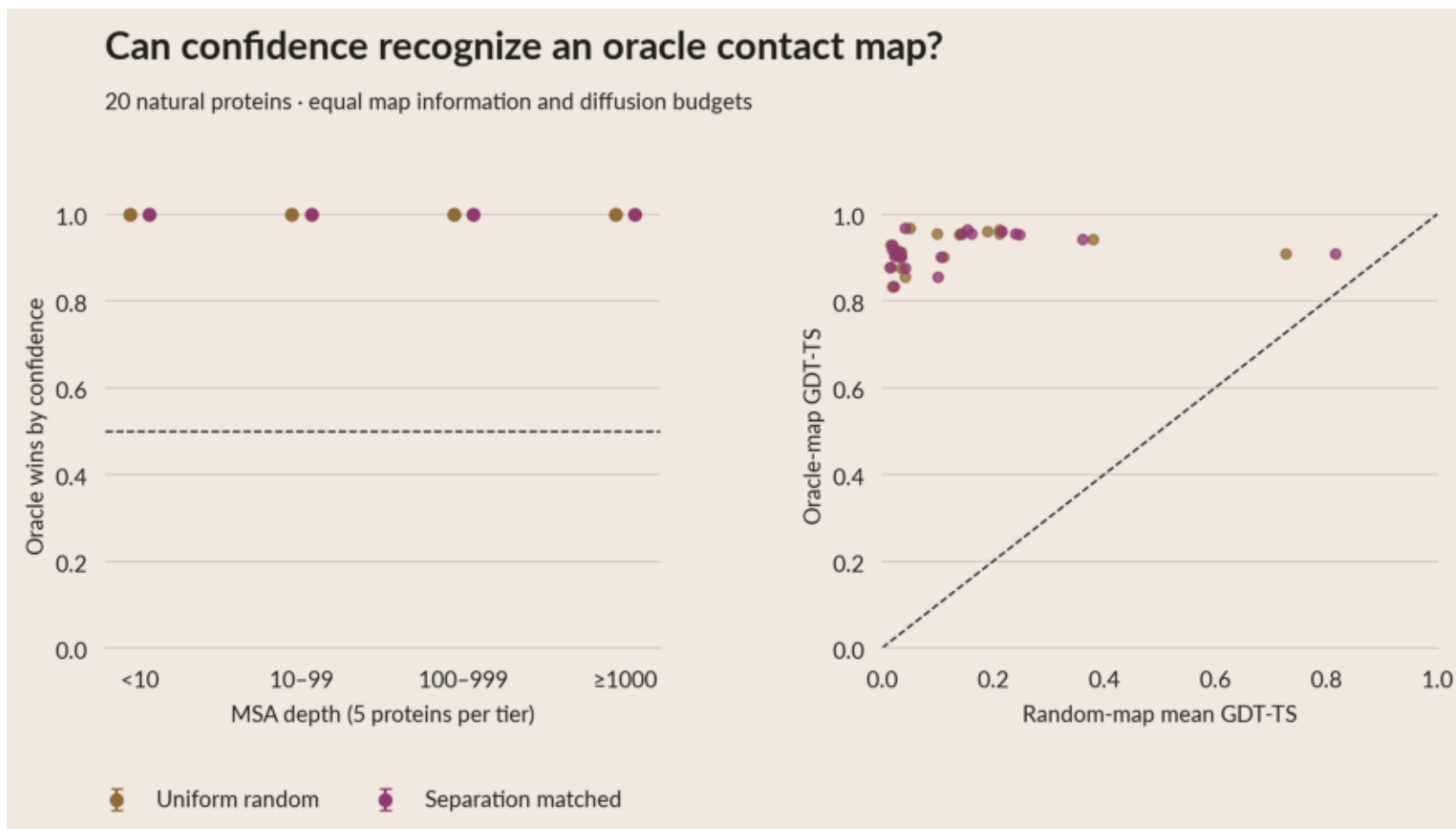
02_oracle_lddt.png

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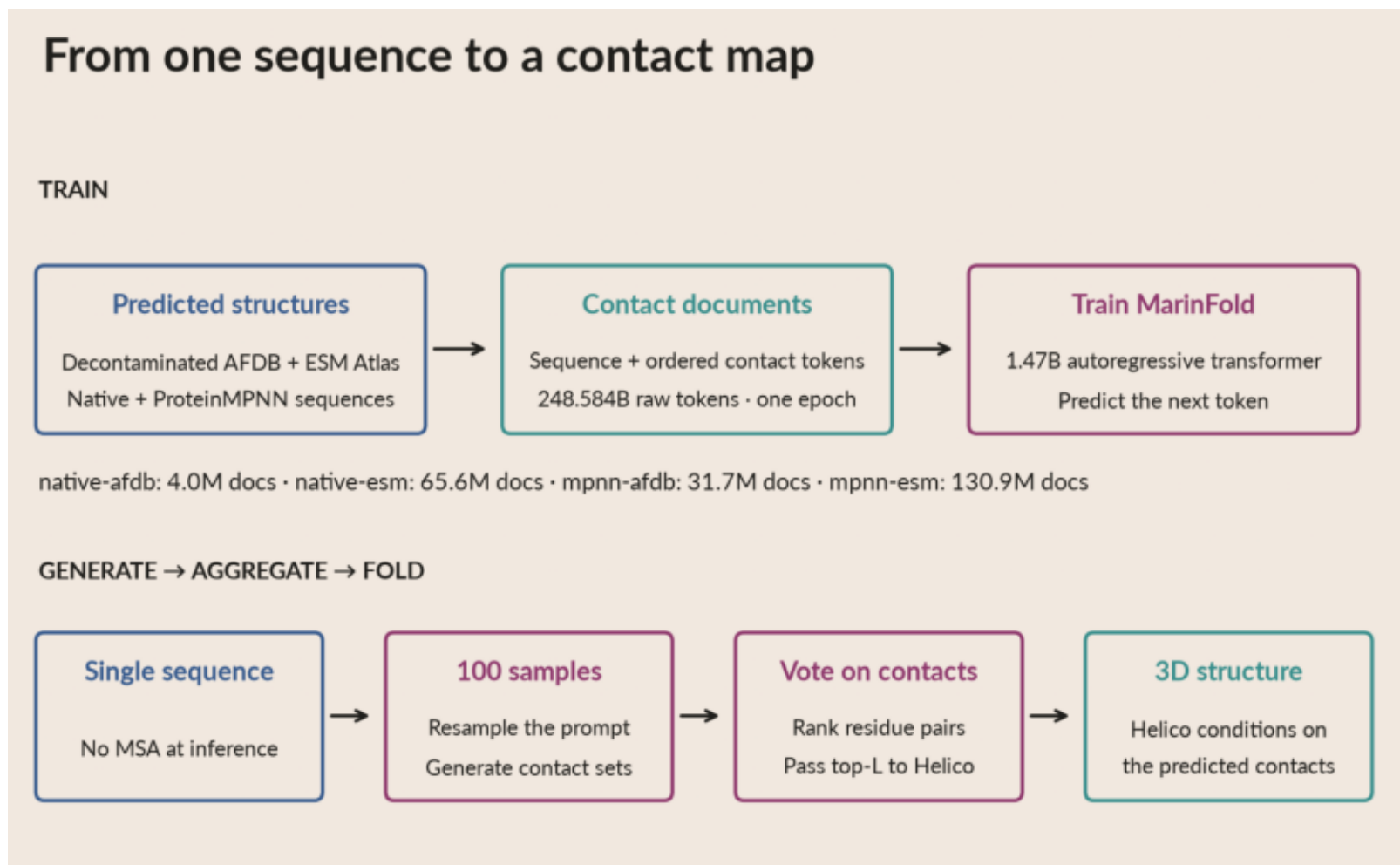
02b_confidence.png

Twenty natural proteins, selected before inference (five per MSA tier). Each oracle is compared with five uniform and five separation-matched random maps; unknown mask, positive/negative counts, and three-sample diffusion budgets are identical. Select each map's structure by Helico ranking_score. Confidence wins are paired within protein; ties count half. Intervals bootstrap proteins, not samples. Random maps are weak negatives; this does not establish ranking of plausible alternative folds.



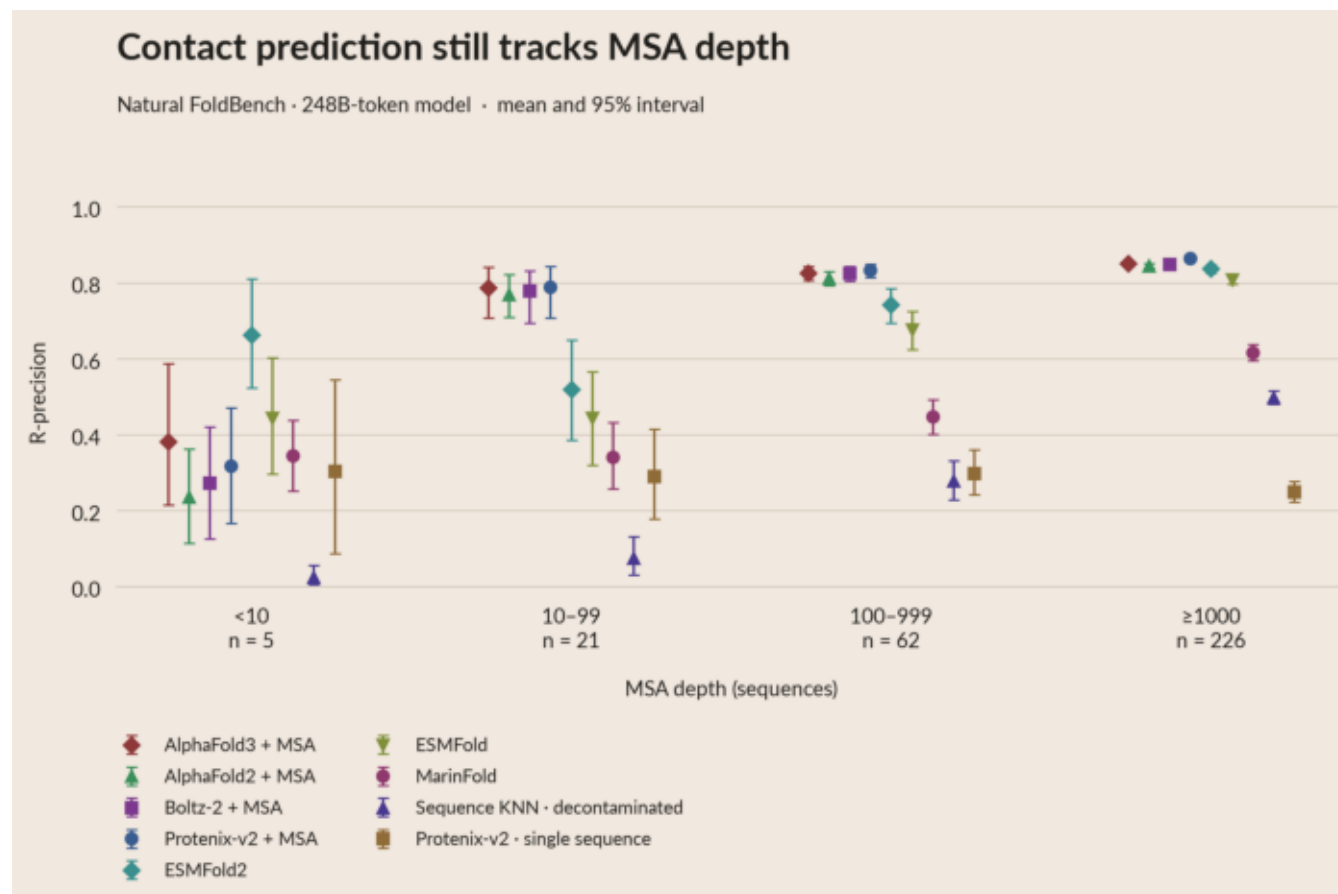
03_method.png

Schematic; arrow widths do not encode amounts. Exp277 step 266,344: a scratch-trained 1.47B-parameter Qwen3, one epoch over 232,090,905 contact documents, totaling 248,583,762,834 raw tokens. Native and ProteinMPNN-redesigned sequences use AFDB / ESM Atlas source structures from the decontaminated corpus. Raw tokens differ from padded training slots.



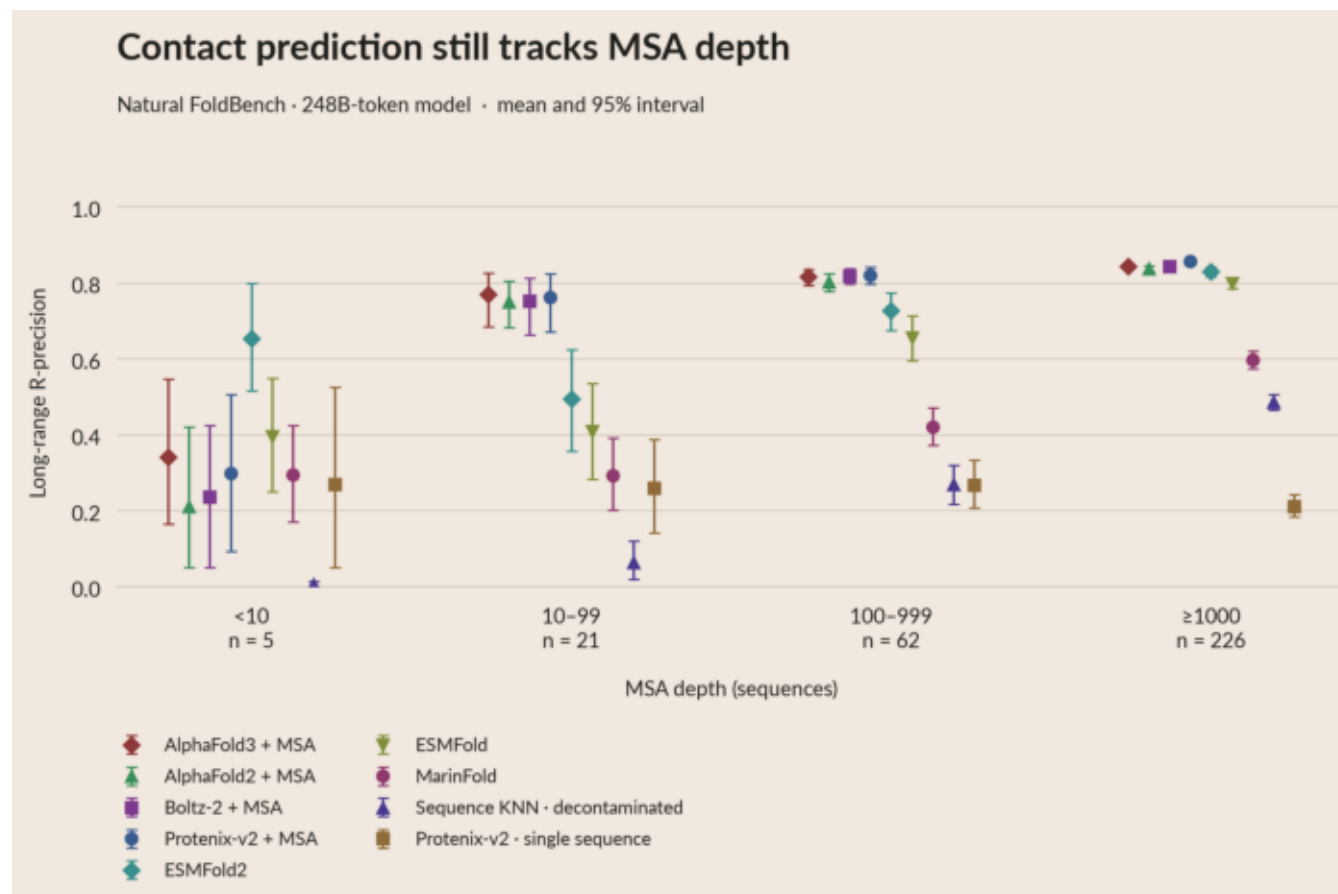
04_contacts.png

Exp277 step 266,344, the 248.584B-raw-token model. All-range R-precision on 314 natural monomers (97 validation + 217 test); 100 resampled rollouts and vote ranking. R is the true-contact count in the resolved candidate universe. KNN indexes the native decontaminated corpus, not the additional redesign sequences. Only five natural proteins have depth <10. The menu also exposes long-range R-precision and separate split/design views. AlphaFold2/3 and Boltz-2 use the same archived MSA queries and alignments underlying these depth bins, with templates disabled and protein-chain-only inputs. Confidence selects among five AF2 pTM models (three recycles), 25 AF3 samples (five seeds, ten recycles), or 25 Boltz-2 samples (one seed, ten recycles). These are controlled shared-MSA runs, not the models' full default search pipelines.



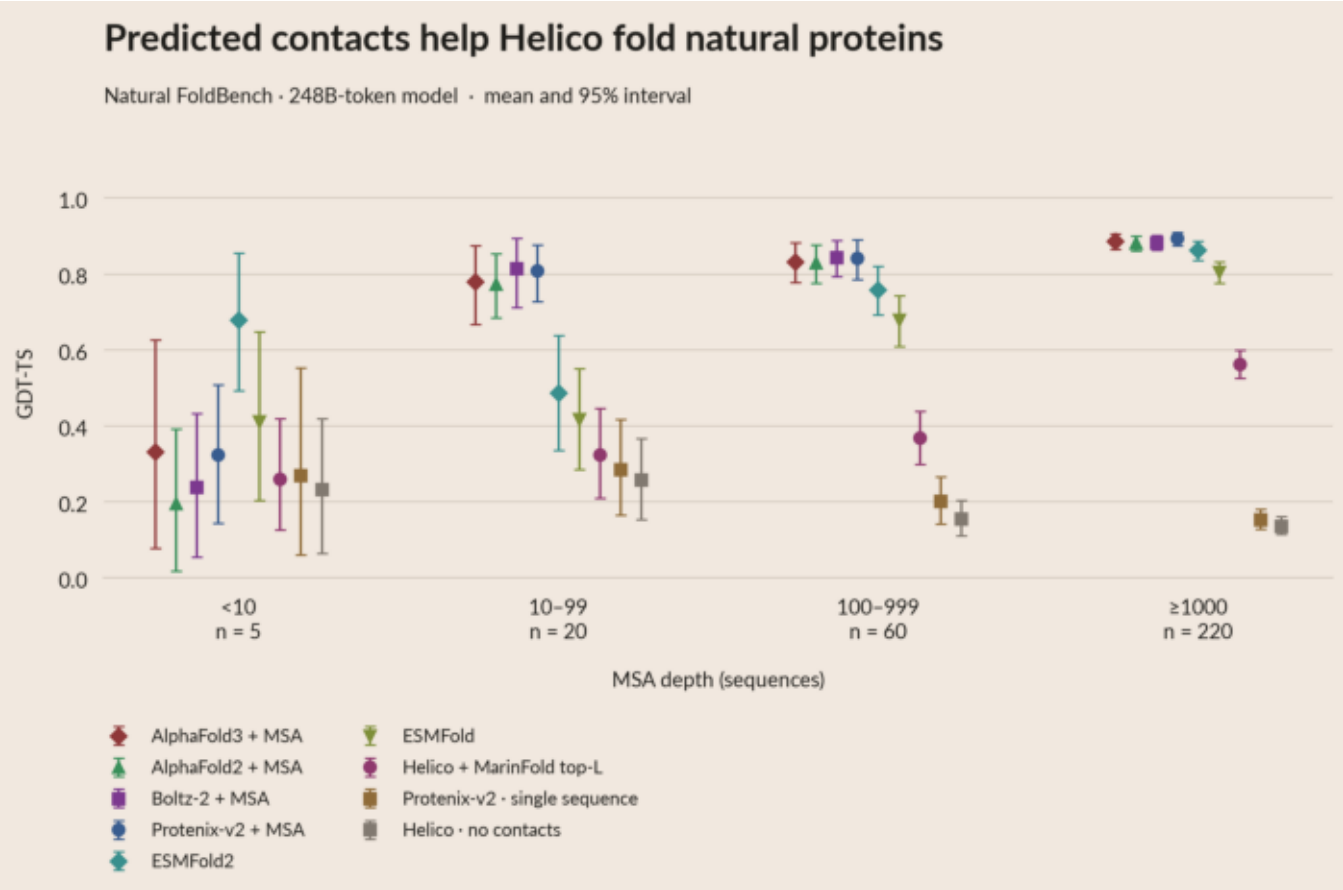
04_contacts_long.png

Exp277 step 266,344, the 248.584B-raw-token model. All-range R-precision on 314 natural monomers (97 validation + 217 test); 100 resampled rollouts and vote ranking. R is the true-contact count in the resolved candidate universe. KNN indexes the native decontaminated corpus, not the additional redesign sequences. Only five natural proteins have depth <10. The menu also exposes long-range R-precision and separate split/design views. AlphaFold2/3 and Boltz-2 use the same archived MSA queries and alignments underlying these depth bins, with templates disabled and protein-chain-only inputs. Confidence selects among five AF2 pTM models (three recycles), 25 AF3 samples (five seeds, ten recycles), or 25 Boltz-2 samples (one seed, ten recycles). These are controlled shared-MSA runs, not the models' full default search pipelines.



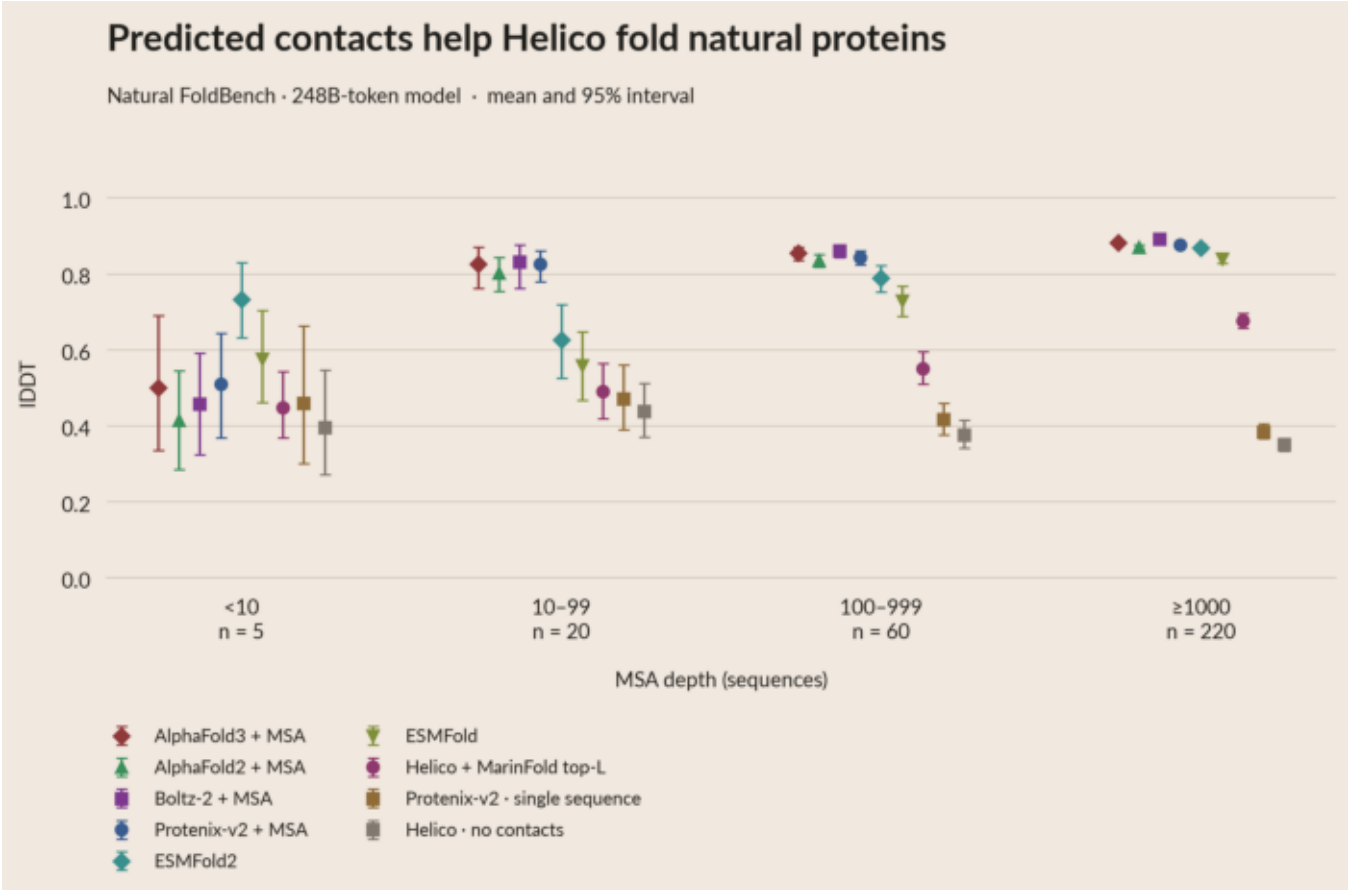
05_folding.png

The 248B-token model through Helico: request the top-L predicted contacts, generate three diffusion samples, select by highest Helico confidence at that fixed cut; no MSA. Helico drops pairs closer than six residues after coordinate mapping; requested and effective counts are saved. Same 305 natural proteins in every arm (95 validation + 210 test). GDT-TS and IDDT are separate metrics. See coverage.csv for exclusions. AlphaFold2/3 and Boltz-2 use the same archived MSA queries and alignments underlying these depth bins, with templates disabled and protein-chain-only inputs. Confidence selects among five AF2 pTM models (three recycles), 25 AF3 samples (five seeds, ten recycles), or 25 Boltz-2 samples (one seed, ten recycles). These are controlled shared-MSA runs, not the models' full default search pipelines.



05_folding_iddt.png

The 248B-token model through Helico: request the top-L predicted contacts, generate three diffusion samples, select by highest Helico confidence at that fixed cut; no MSA. Helico drops pairs closer than six residues after coordinate mapping; requested and effective counts are saved. Same 305 natural proteins in every arm (95 validation + 210 test). GDT-TS and IDDT are separate metrics. See coverage.csv for exclusions. AlphaFold2/3 and Boltz-2 use the same archived MSA queries and alignments underlying these depth bins, with templates disabled and protein-chain-only inputs. Confidence selects among five AF2 pTM models (three recycles), 25 AF3 samples (five seeds, ten recycles), or 25 Boltz-2 samples (one seed, ten recycles). These are controlled shared-MSA runs, not the models' full default search pipelines.



06_sampling.png

Exp277 step 266,344, 314 natural proteins (97 validation + 217 test), 100 iid rollouts per protein. Consensus and oracle use the same sample pool within each protein. Individual maps are ranked by emission order; short maps retain denominator R, and malformed or unfinished individual maps score zero. The diagnostic consensus uses all parsed maps, following exp321. Oracle selection uses ground truth. This measures contact accuracy, not distinct folds. Scatter colors identify MSA tiers in the interactive view.

